

ELEC3202: Coursework: PC1D Solar Cell Simulation Exercise 2026

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(The space for each task should be expanded as required, but the report overall should not exceed 10 pages)

Task 1: Dark and illuminated I-V characteristics

(a)

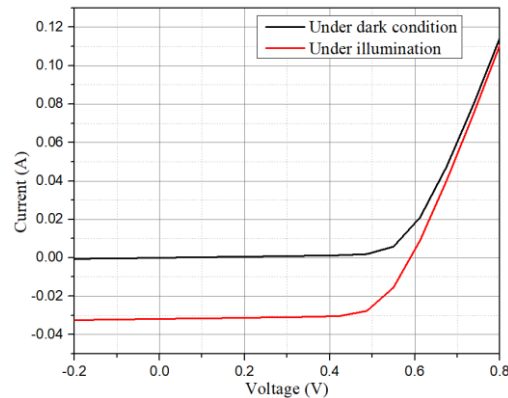


Figure 1 Base IV characteristics under dark and illuminated conditions

(b)

The total current in a solar cell is the sum of the dark current and the current generated by the light.

$$I_{total} = I_{dark} - I_L$$

The black curve represents the dark current (I_{dark}). When the solar cell is illuminated by light, it creates e^-h^+ pairs, which results in a light-generated current (I_L). This light current flows in the opposite direction to the forward-bias current of the diode [1].

EXCITATION

Excitation modified from one-sun.exc
Excitation mode: Transient, 16 timesteps
Temperature: 25°C
Base circuit: Sweep from -0.2 to 0.8 V
Collector circuit: Zero
Light sources disabled

RESULTS

Short-circuit Ib: 1.825e-9 amps

Open-circuit Vb: -6.111e-7 volts

Figure 2 Excitation under dark conditions

Since I_L is subtracted from the dark current, the entire curve is shifted downward along the y-axis. At $V=0$ (short circuit): In the dark, the current is 0 and under illumination, the current is approximately **-0.0318 A**. This is the value of the short-circuit current (I_{sc}), which is directly proportional to the intensity of light. The point where the red line crosses the x-axis ($I=0$) is the V_{oc} , which is **0.5895 V**.

EXCITATION
Excitation modified from one-sun.exc
Excitation mode: Transient, 16 timesteps
Temperature: 25°C
Base circuit: Sweep from -0.2 to 0.8 V
Collector circuit: Zero
Primary light source enabled
Constant intensity: 0.1 W cm⁻²
Spectrum from am15g.spc
Secondary light source disabled

RESULTS
Short-circuit Ib: -0.0318 amps
Max base power out: 0.0139 watts
Open-circuit Vb: 0.5895 volts

Figure 3 Excitation under illuminated conditions

(c)

$$\text{Fill factor} = \frac{P_{max}}{I_{sc} V_{oc}} \quad \text{Fill factor} = \frac{0.0139}{0.0318 \times 0.5895} = 0.741$$

From the simulation, it is given that the constant intensity is 0.1 W cm⁻², therefore, $P_{light} = 0.1 \text{ W}$.

Power conversion efficiency (PCE):

$$PCE = \frac{P_{max}}{P_{light}} = \frac{0.0139}{0.1} = 0.139 = 13.9\%$$

A PCE of 13.9% for a photovoltaic cell is considered modest to low by modern standards, but it can be excellent for specialized technologies like organic and thin-film cells [2].

Task 2: Varying illumination intensity

(a)

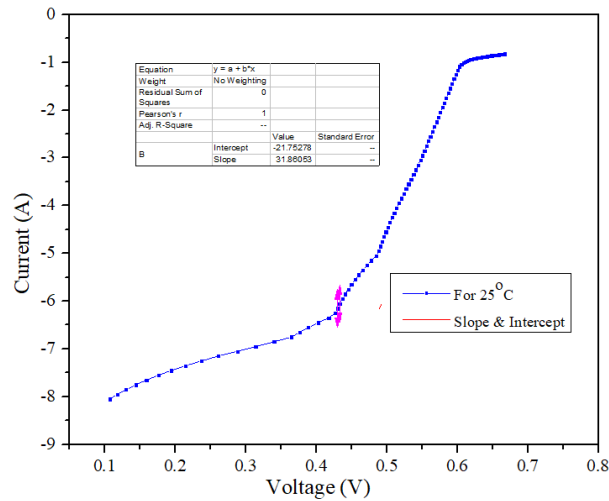


Figure 4 Ideality factor at 25 °C

From the $\ln(-I_{sc})$ Vs V_{oc} graph, we get a slope of 31.86053. We know that $\ln(-I_{sc}) = \frac{qV_{oc}}{nkT} + \ln I_o$

$$\text{slope} = \frac{q}{nkT} = 31.86053 \quad n = \frac{1.602 \times 10^{-19}}{31.86053 \times 1.380 \times 10^{-23} \times 298} = 1.22163$$

From the intercept of the slope on the y-axis:

$$\ln I_o = -21.275878 \quad I_o = 5.75 * 10^{-10} A$$

This value of I_o suggests that the simulated photovoltaic cell works as a commercialised standard solar cell. The lower the value of I_o the higher the V_{oc} , which represents an increase in efficiency [1].

(b) Changes at 50 °C

EXCITATION
Excitation modified from one-sun.exc
Excitation mode: Transient, 16 timesteps
Temperature: 50°C
Base circuit: Sweep from -0.2 to 0.8 V
Collector circuit: Zero
Primary light source enabled
Constant intensity: 0.1 W cm⁻²
Spectrum from am15g.spc
Secondary light source disabled

RESULTS
Short-circuit Ib: -0.0319 amps
Max base power out: 0.0121 watts
Open-circuit Vb: 0.5344 volts

Figure 5 Excitation at 50 °C

- $Fill\ factor = \frac{0.0121}{0.0319 * 0.5344} = 0.709$ $PCE = \frac{0.0121}{0.1} = 0.121 = 12.1\%$
- $\ln I_o = -18.73182$ $I_o = 7.32 * 10^{-9} A$

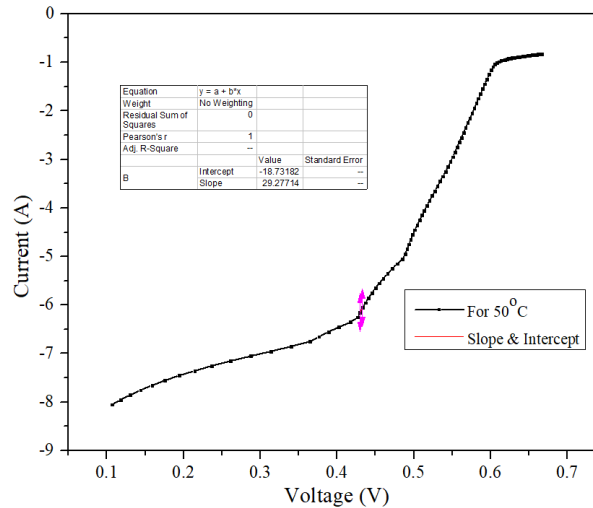


Figure 6 Ideality factor at 50 degrees

With specialized cooling and grid design, c-Si cells can technically handle much higher intensities, often upto 30 W cm⁻².

To design a system with passive cooling, we should not exceed 0.2 to 0.3 W cm⁻² and with active cooling, upto 10 to 20 W cm⁻². To operate at higher intensities, we have to use “low series resistance” cells specifically designed with denser top-grid to handle the massive current [3].

Table 1 Quick batch results

PrInsyTR1	PrInsyTR2	BaseVoc	BaseIsc
0.001	0.001	0.10653	-0.00032
1.05358	1.05358	0.652609	-0.33518
2.10616	2.10616	0.669253	-0.4234
3.15874	3.15874	0.678515	-0.4368
4.21132	4.21132	0.684813	-0.44389
5.26389	5.26389	0.689518	-0.44865
6.31647	6.31647	0.693232	-0.45217
7.36905	7.36905	0.696275	-0.45496
8.42163	8.42163	0.698836	-0.45724
9.47421	9.47421	0.701037	-0.45916
10.5268	10.5268	0.702957	-0.46082
11.5794	11.5794	0.704655	-0.46227
12.6319	12.6319	0.706174	-0.46355
13.6845	13.6845	0.707543	-0.46471
14.7371	14.7371	0.708787	-0.46575
15.7897	15.7897	0.709926	-0.4667
16.8423	16.8423	0.710975	-0.46757
17.8948	17.8948	0.711945	-0.46837
18.9474	18.9474	0.712846	-0.46911
20	20	0.713687	-0.46981

Table 2 Changes in device parameters when the temperature is increased

Parameter	At 25 °C	At 50 °C
I_{sc}	-0.0318 A	-0.0319 A
V_{oc}	0.5895 V	0.5344 V
P_{max}	0.0139 W	0.0121 W
Fill factor	0.741	0.709
PCE	13.9%	12.1 %
Ideality factor	1.22	1.22
I_o	$5.75 * 10^{-10}$ A	$7.32 * 10^{-9}$ A

Observations:

- The band gap of a semiconductor is not a fixed value; it decreases as the temperature rises. As the crystal lattice vibrates more intensely, the interatomic spacing changes, reducing the energy required to excite an electron from the valence band to the conduction band [4].
- A narrower band gap allows the cell to absorb more lower-energy photons, which results in a slight increase in the I_{sc} . It also increases the intrinsic carrier concentration, which leads to a much larger increase in the dark saturation current I_o [4].
- Since I_o increases exponentially, V_{oc} shrinks, and the P_{max} decreases because of the decrease in V_{oc} , the Fill Factor also decreases, and the efficiency goes down.
- The bandgap of silicon is sensitive to heat. For every °C increase above 25°C, the V_{oc} drops by about 0.4%. Without active cooling, intensities above 0.5 W/cm² will quickly heat the cell to temperatures that permanently damage the polymer packaging [4].

Task 3: Front surface optimization

(a)

Surface texturing:

The major change in the device architecture is the shift to a high-efficiency light-trapping scheme. This is done using an online tool called “Wafer ray tracer”. Both the front and rear surfaces of the cell have been textured with randomly periodic upright pyramids. This morphology is designed to increase the optical path length of incident photons through multiple internal reflections [5].

In an attempt to decrease the size of the substrate to reduce bulk recombination and decrease material costs, it is observed that the efficiency of the cell also decreases. It happens because the optical path gets decreased and the carriers have less space to form $e^- - h^+$ pairs, so it absorbs fewer photons, particularly of infrared wavelength, as they require a large space to get absorbed. Thinning the wafer may reduce the bulk recombination, but in turn, it also reduces the efficiency.

Thin Film Passivation and Reflectivity

In order to complement the physical texturing, special thin-film finishes were used to increase the optical and electrical characteristics of the interfaces:

- **Front Surface:** A DLAR (Double Layer Anti-reflection) coating is made.
 1. A layer of SiN_x 75 nm in size has been deposited. This film serves as a dual-purpose coating, which provides essential surface passivation to reduce carrier recombination while also acting as an Anti-Reflective Coating (ARC) to minimize front-surface shading [6] [7].
 2. A 20 nm layer of Al_2O_3 is applied as a passivation layer, which also acts as an ARC.
- **Rear surface:**
 1. A layer of SiO_2 100 nm in size has been incorporated at the back of the cell. This layer serves as a rear dielectric reflector such that long-wavelength photons penetrating to the rear of the cell are reflected back into the silicon substrate to be absorbed a second time.
 2. A 20 nm layer of Al is applied, which acts as a rear reflector to bounce off the unabsorbed light back into the substrate [7][8].



Figure 7 Reflectance, Absorbance, and Transmittance of the Solar Cell

- 1) **Absorbance (A-green line):** The absorbance represents the fraction of light used for potential e^-h^+ pair generation.
 - Optimal Range (450 nm – 950 nm): In this range, absorbance is nearly 1.0 (100%). It represents that the silicon is thick enough, and the energy is high enough for nearly all photons to be absorbed.
 - There is a sharp drop-off starting around 1000 nm. Silicon has an indirect band gap of 1.12 eV, which corresponds to a wavelength of approximately 1100 nm. Photons with wavelengths longer than this, i.e. having low enough energy that they could not excite the e^- , cause the absorbance to plummet to zero.
- 2) **Reflectance (R-red line):** Reflectance represents the light that "bounces" off the surface and is lost before it can enter the cell.
 1. Low Reflection: The reflectance is kept near 0 in the range of 450 nm – 1000 nm. This indicates the presence of an Anti-Reflective Coating (ARC) and surface texturing, which helps in light trapping.
 2. The rise in reflectance at both the very low wavelength and very high wavelength ends shows where the ARC becomes less effective.
- 3) **Transmittance (T-blue line):** Transmittance shows the light that passes straight through the wafer without being absorbed.
 - In the range of 300 nm to 950 nm, transmittance is zero. The silicon is effectively opaque because it absorbs all the light that isn't reflected.
 - Once the wavelength exceeds the band gap energy, the silicon becomes transparent. These low-energy photons simply pass through the wafer, which is why the blue line climbs sharply as the green line falls.

INPUTS

■ Illumination

Angle of incidence
 Zenith, θ °
 → Switch to isotropic

Spectrum

Sunlight
▼
AM1.5g [Gue95]
▼

 → Set standard test conditions

■ Surface morphology

Side	Morphology	Periodicity	Angle (°)	Height (μm)	Width (μm)
Front	Upright pyramids ▼	Random ▼	54.74	3.536	5
Rear	Upright pyramids ▼	Random ▼	54.74	3.536	5

■ Layer materials

Layer	Thickness	Material
Surrounds		Air ▼ [] ▼
x Front film	75 nm	SiNx ▼ PECVD [Bak11] ▼
x Front film	20 nm	Al2O3 ▼ ALD on glass [Kum09] ▼
Substrate	300 μm	Si ▼ Crystalline, 300 K [Gre08] ▼
x Rear film	100 nm	SiO2 ▼ Thermal [Pal85e] ▼
x Rear film	20 nm	Al ▼ Pure [Pal85b] ▼

+ Add front film + Add rear film
 x Clear all front films x Clear all rear films

↕ Flip material stack
 ≡ Symmetrical structure

→ Switch to rear reflector
 + Add detached reflector

■ Options

Minimum wavelength 300 nm
 Maximum wavelength 1200 nm
 Wavelength interval 10 nm
 Calculate gen profile ☐

Number of rays per run 5000
 Max total rays 50000
 Max bounces per ray 1000
 Intensity limit 0.01 %
 Minimise error in J_G ☒
 Significant figures 4

Figure 8 Inputs in wafer ray tracer

OUTPUTS			
Ray tracing complete. Total rays traced: 50000.			
■ Photon current densities			
		Mean $\pm$ 95% CI (mA/cm ²)	Fraction of J_{inc} (%)
Incident	J_{inc}	46.29 $\pm$ 4.389e-15	100.0
Lost			
Reflected – external	$J_{R,ext}$	1.064 $\pm$ 0.04613	2.297
Reflected – escape	$J_{R,esc}$	1.145 $\pm$ 0.03439	2.473
Transmitted	J_T	0.06167 $\pm$ 0.01041	0.1332
Absorbed – front films	$J_{A,F}$	0.5501 $\pm$ 0.003041	1.188
Absorbed – rear films	$J_{A,R}$	0.9856 $\pm$ 0.01754	2.129
Absorbed – detached reflector	$J_{A, det refl}$	– $\pm$ –	–
Photogeneration (absorbed)	J_G	42.43 $\pm$ 0.03924	91.66
Remainder		0.05649 $\pm$ 0.008825	0.1220

Figure 9 Percentage of Reflectance, Absorbance, and Transmittance

Observation:

Different ways of texturing, such as upright and inverted pyramids, are simulated, and it is observed that the inverted pyramids give superior results for the same structure compared to upright pyramids. The absorption rate for inverted pyramids is 92.54% as compared to 91.66 % of upright pyramids. But in this experiment, I am using upright pyramids for texturing because of their easy and low-cost fabrication procedure.

(b)

Table 3 Comparison of Wafer Ray Tracer results with untextured & uncoated solar cell

Parameter	Textured and coated	Untextured and uncoated
I_{sc}	–0.0347A	–0.0219 A
V_{oc}	0.5922 V	0.5814 V
P_{max}	0.0152 W	0.009542 W
PCE	15.2%	9 %

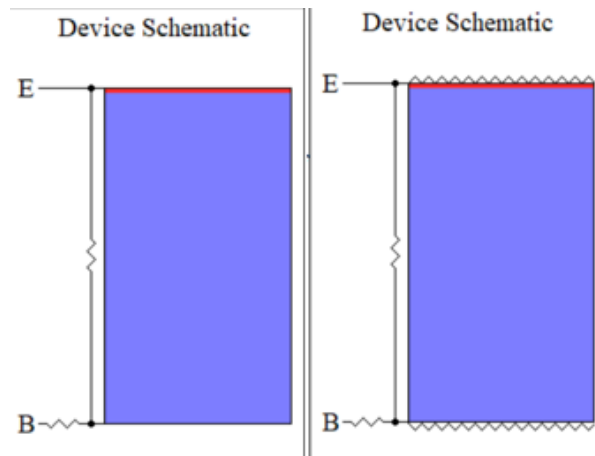


Figure 10 (a) Untextured and uncoated (b) Textured and coated

Observations:

- Increase in Short-Circuit Current:** The I_{sc} is the current where the voltage is zero. The solar cell that is textured and coated comparably produces more current. This is because texturing creates microscopic pyramids on the surface that trap light, and the anti-reflective coatings (ARC) reduce the amount of light reflected off the cell, allowing more photons to be absorbed and converted into $e^- - h^+$ pairs [1].

- ii. **Open-Circuit Voltage:** The V_{oc} is the point where the current crosses zero on the x-axis. Both cells have a V_{oc} around 0.6 V. While texturing has a much stronger influence on current, the effect on voltage is generally much less since V_{oc} depends more on the bandgap and recombination rates of the material used [1].
- iii. **Fill Factor and Power Output:** Since the area under the curve of the IV characteristic represents power $P = I \cdot V$, the black curve encloses a much larger area than the red curve, indicating a higher P_{max} and higher overall efficiency [1].

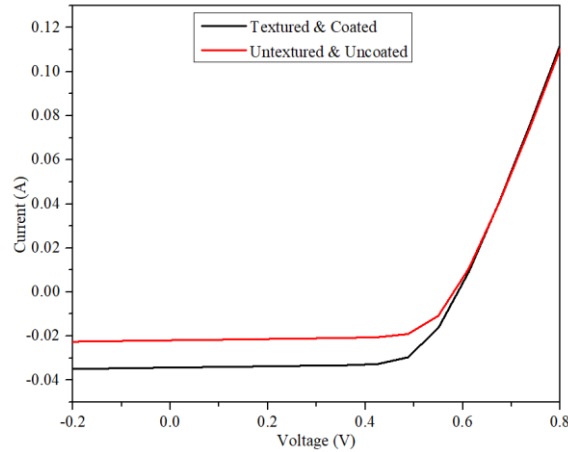


Figure 11 Comparison of IV characteristics of textured & coated and untextured & uncoated photovoltaic cells

- **Why Texturing and Coating Matter**

Texturing: By creating a rough plane using pyramidal structures, it ensures that the light that reflects off the surface and bounces off another pyramidal structure gives the light a second chance to be absorbed. This also increases the path length of light within the silicon (light trapping) [5].

ARC: A thin layer of anti-reflecting material, e.g. SiN_x , uses destructive interference to cancel out reflected light, making the cell look dark blue or black instead of shiny grey.

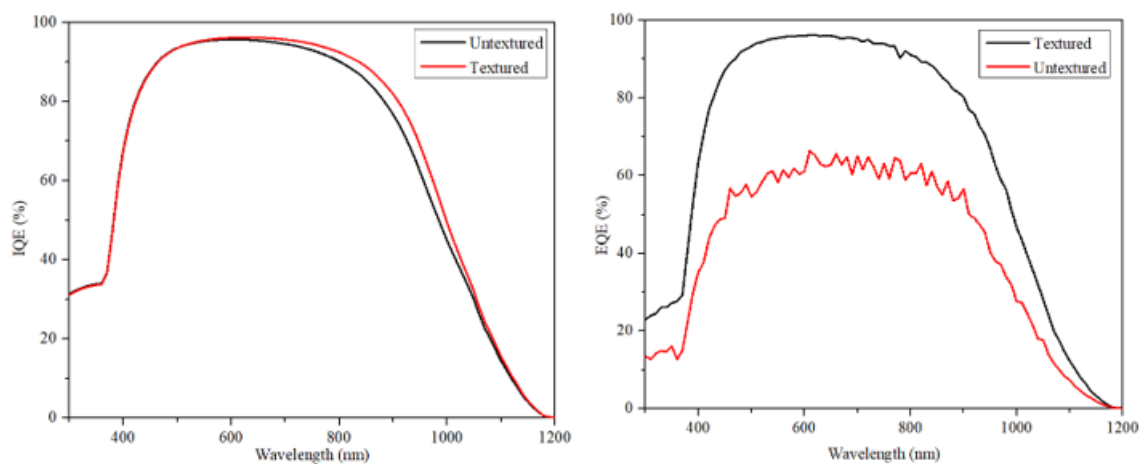


Figure 12 Comparison of the quantum efficiency of textured and untextured solar cells

A. External Quantum Efficiency (EQE): EQE measures the ratio of collected charge carriers to the number of photons incident on the solar cell surface.

- **Observation:**

The "Textured" curve is significantly higher than the "Untextured" curve across almost the entire spectrum. This shows that an untextured (flat) silicon surface is highly reflective, acting like a mirror and bouncing away a huge chunk, about 30% of incoming light. Texturing reduces this reflection by giving photons multiple chances to enter the cell [6] [9].

- The massive gap between the red and black lines represents the reduction in reflection losses.

B. Internal Quantum Efficiency (IQE): IQE measures the ratio of collected charge carriers to the number of photons that are actually absorbed, ignoring those that are reflected away. The graph shows that the two cells are nearly identical in internal performance [9].

- **Observation:**

The red and black curves almost perfectly overlap, especially in the wavelength range of 400 nm to 800 nm. This indicates that the diffusion length, carrier lifetime, and surface passivation are the same for both semiconductor cells.

- It can be observed that the textured cell performs slightly better at high wavelengths (>900 nm). This is because texturing increases the optical path length; photons travel at an angle, making the cell appear thicker to the light and increasing the probability of absorption for near-infrared photons [5][7].

(c) Although being able to reduce reflectance is undeniably a good thing, to design a really effective solar cell, the top surface should also be efficient in its carrier collection, electrical conduction, and protection capabilities [10].

The top-surface design of a solar cell is determined by the following factors:

1. Front contact shadowing:

- i. Reduced thickness and increased frequency of the fingers enhance the resistance of the emitter that electrons have to cover. Preferably, the fingers must be long and thin to offer minimal resistance and allow considerable light to pass through [10].
- ii. This is maximized using techniques such as Buried contact or Silver screen printing [3].

2. Surface recombination:

It serves as an enormous source of traps for e^- and h^+ because of the existence of the dangling bonds located on the silicon surface in the top surface. Thus, the passivation of such bonds is done by applying a thin material such as SiO_2 . These layers must be insulated with electricity, so that they can work [10][3].

3. Lateral series resistance:

An electron has to travel through the top layer of the silicon before it reaches a metal finger. To reduce this resistance, a selective emitter approach is used, where heavy doping is used only under the metal contacts and light doping in the sunny areas between fingers to balance low resistance with high carrier lifetime [10][9].

4. Transparency of the TCO (Transparent Conducting Oxide):

In other films, such as SHJ, a TCO layer is applied throughout the surface to enhance conductivity, although it absorbs more photons, which are not supposed to be absorbed into the silicon [7][9].

Conclusion:

We know that the best PCE of an AL-BSF is approximately 19.5 % under lab conditions. The structure that we have made is based on a commercial model, so a PCE of 15.2 % can be considered very good [2][4][9].

References

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